

# Лабораторная работа №2

Простые сети в GNS3. Анализ трафика

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1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
3. Выводы по проделанной работе

## 1. 1. Цели и задачи работы

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## 1.1 Цель лабораторной работы

Получение практических навыков моделирования простых сетей в GNS3, настройки IP-адресации конечных устройств и маршрутизаторов, а также анализа сетевого трафика с помощью Wireshark.

## 2. 2. Процесс выполнения лабораторной работы

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## 2.1 Создаю простую сеть на базе коммутатора

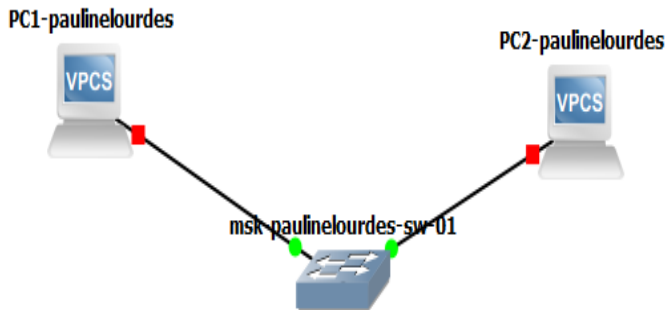


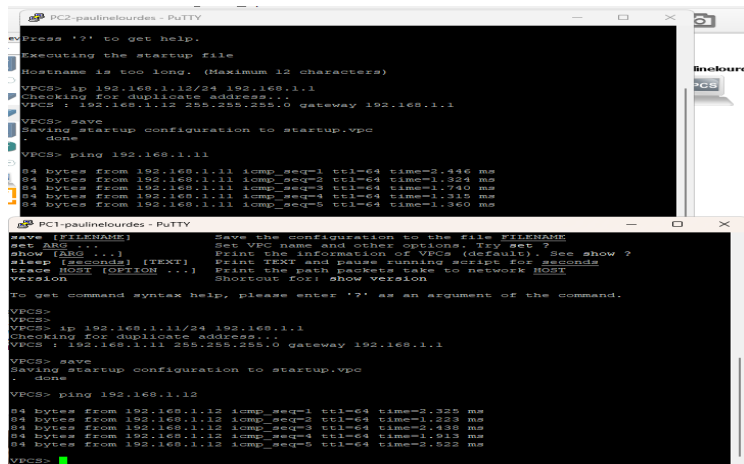
Рисунок 1: Топология простейшей сети на базе Ethernet-коммутатора

## 2.2 Просматриваю команды VPCS

```
VPCS> ?  
?  
arp          Shortcut for: show arp. Show arp table  
clear ARG    Clear IPv4/IPv6, arp/neighbor cache, command history  
dhcp [OPTION] Shortcut for: ip dhcp. Get IPv4 address via DHCP  
disconnect   Exit the telnet session (daemon mode)  
echo TEXT    Display TEXT in output. See also set echo ?  
help         Print help  
history      Shortcut for: show history. List the command history  
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?  
load [FILENAME] Load the configuration/script from the file FILENAME  
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?  
quit        Quit program  
relay ARG ... Configure packet relay between UDP ports. See relay ?  
rlogin [ip] port Telnet to port on host at ip (relative to host PC)  
save [FILENAME] Save the configuration to the file FILENAME  
set ARG ...   Set VPC name and other options. Try set ?  
show [ARG ...] Print the information of VPCs (default). See show ?  
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds  
trace HOST [OPTION ...] Print the path packets take to network HOST  
version      Shortcut for: show version  
  
To get command syntax help, please enter '?' as an argument of the command.  
VPCS>
```

Рисунок 2: Просмотр справки по доступным командам VPCS

## 2.3 Настраиваю IP-адреса и проверяю связь



The image shows two PuTTY terminal windows. The top window, titled 'PC2-paulinelourdes - PuTTY', shows the VPCS configuration for PC2. The bottom window, titled 'PC1-paulinelourdes - PuTTY', shows the VPCS configuration for PC1 and a successful ping test to PC2.

```
PC2-paulinelourdes - PuTTY
Press '?' to get help.
Executing the startup file
Hostname is too long. (Maximum 12 characters)
VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1
VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> ping 192.168.1.11
64 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.446 ms
64 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.324 ms
64 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.740 ms
64 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.315 ms
64 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.360 ms

PC1-paulinelourdes - PuTTY
save [FILENAME]      Save the configuration to the file FILENAME
set ARG ...          Set VPC name and other options. Try set ?
show [ARG ...]       Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version              Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.

VPCS>
VPCS>
VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1
VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> ping 192.168.1.12
64 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=2.325 ms
64 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=1.223 ms
64 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=2.438 ms
64 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=1.913 ms
64 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=2.522 ms
VPCS>
```

Рисунок 3: Настройка IP-адресов и проверка связи между PC1 и PC2

## 2.4 Запускаю захват трафика в GNS3

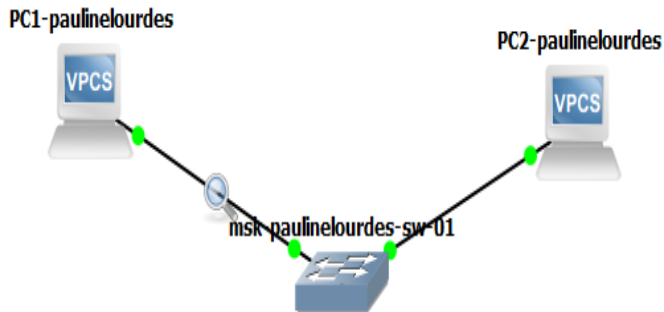


Рисунок 4: Запуск захвата трафика между PC1 и коммутатором

## 2.5 Анализирую ARP-сообщения

| No. | Time     | Source           | Destination | Protocol | Length | Info                                      |
|-----|----------|------------------|-------------|----------|--------|-------------------------------------------|
| 1   | 0.000000 | ::               | ff02::1     | ICMPv6   | 62     | Router Solicitation                       |
| 2   | 0.000353 | ::               | ff02::1     | ICMPv6   | 62     | Router Solicitation                       |
| 3   | 0.051683 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 4   | 0.052223 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 5   | 1.051437 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 6   | 1.051918 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 7   | 2.051757 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 8   | 2.052283 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>[Time shift for this packet: 0.000000000 seconds]</div> <div>[Time delta from previous captured frame: 540.000 microseconds]</div> <div>[Time delta from previous displayed frame: 540.000 microseconds]</div> <div>[Time since reference or first frame: 52.223000 milliseconds]</div> <div>Frame Number: 4</div> <div>Frame Length: 64 bytes (512 bits)</div> <div>Capture Length: 64 bytes (512 bits)</div> <div>[Frame is marked: False]</div> <div>[Frame is ignored: False]</div> <div>[Protocols in frame: ethertype:arp]</div> <div>Character encoding: ASCII (0)</div> <div>[Coloring Rule Name: ARP]</div> <div>[Coloring Rule String: arp]</div> <div>▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)</div> <div>&gt; Destination: Broadcast (ff:ff:ff:ff:ff:ff)</div> <div>&gt; Source: Private_66:68:01 (00:50:79:66:68:01)</div> <div>Type: ARP (0x0806)</div> <div>[Stream index: 3]</div> <div>Padding: 00000000000000000000000000000000</div> <div>Frame check sequence: 0x00000000 [unverified]</div> <div>[FCS Status: Unverified]</div> <div>▼ Address Resolution Protocol (request/gratuitous ARP)</div> <div>Hardware type: Ethernet (1)</div> <div>Protocol type: IPv4 (0x0800)</div> <div>Hardware size: 6</div> <div>Protocol size: 4</div> <div>Opcode: request (1)</div> <div>[Is gratuitous: True]</div> <div>Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)</div> <div>Sender IP address: 192.168.1.12</div> <div>Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)</div> <div>Target IP address: 192.168.1.12</div> | <div>0000 ff ff ff ff ff ff 00 50 79 66 68 01 08 06 00 01 .....P yfh---</div> <div>0010 08 00 06 04 00 01 00 50 79 66 68 01 c0 a8 01 8c .....P yfh---</div> <div>0020 ff ff ff ff ff ff c0 a8 01 8c 00 00 00 00 00 00 .....P yfh---</div> <div>0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....P yfh---</div> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Рисунок 5: Анализ широковещательных ARP-сообщений в Wireshark

## 2.6 Просматриваю параметры команды ping

```
VPCS> ping ?  
  
ping HOST [OPTION ...]  
Ping the network HOST. HOST can be an ip address or name  
Options:  
-1          ICMP mode, default  
-2          UDP mode  
-3          TCP mode  
-c count   Packet count, default 5  
-D          Set the Don't Fragment bit  
-f FLAG    Tcp header FLAG |C|E|U|A|P|R|S|F|  
              bits |7 6 5 4 3 2 1 0|  
-i ms      Wait ms milliseconds between sending each packet  
-l size    Data size  
-P protocol Use IP protocol in ping packets  
              1 - ICMP (default), 17 - UDP, 6 - TCP  
-p port    Destination port  
-s port    Source port  
-T ttl     Set ttl, default 64  
-t          Send packets until interrupted by Ctrl+C  
-w ms     Wait ms milliseconds to receive the response  
  
Notes: 1. Using names requires DNS to be set.  
       2. Use Ctrl+C to stop the command.
```

Рисунок 6: Просмотр параметров команды ping в VPCS

## 2.7 Выполняю ICMP-проверку

```
VPCS> ping 192.168.1.11 -c 1 -l
```

```
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.443 ms
```

```
VPCS> 
```

Рисунок 7: Одиночный эхо-запрос в ICMP-режиме

## 2.8 Анализирую ICMP-трафик

The image shows a Wireshark packet capture analysis of ICMP traffic. The top pane displays a list of packets, with packet 12 selected. The middle pane shows the packet details for the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

**Packet List:**

| No. | Time       | Source           | Destination      | Protocol | Length | Info                                                           |
|-----|------------|------------------|------------------|----------|--------|----------------------------------------------------------------|
| 4   | 0.052223   | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                      |
| 5   | 1.051437   | Private_66:68:00 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request)                      |
| 6   | 1.051918   | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                      |
| 7   | 2.051757   | Private_66:68:00 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request)                      |
| 8   | 2.052283   | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                      |
| 9   | 254.399614 | Private_66:68:01 | Broadcast        | ARP      | 64     | Who has 192.168.1.11? Tell 192.168.1.12                        |
| 10  | 254.400916 | Private_66:68:00 | Private_66:68:01 | ARP      | 64     | 192.168.1.11 is at 00:50:79:66:68:00                           |
| 12  | 254.402411 | 192.168.1.12     | 192.168.1.11     | ICMP     | 98     | Echo (ping) request id=0x2249, seq=1/256, ttl=64 (reply in 12) |
| 13  | 254.403335 | 192.168.1.11     | 192.168.1.12     | ICMP     | 98     | Echo (ping) reply id=0x2249, seq=1/256, ttl=64 (request in 11) |

**Packet Details (Packet 12):**

- Ethernet II, Src: Private\_66:68:01 (00:50:79:66:68:01), Dst: Private\_66:68:00 (00:50:79:66:68:00)
  - Destination: Private\_66:68:00 (00:50:79:66:68:00)
  - Source: Private\_66:68:01 (00:50:79:66:68:01)
  - Type: IPv4 (0x0800)
  - [Stream index: 4]
- Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
  - 0100 .... = Version: 4
  - .... 0101 = Header Length: 20 bytes (5)
  - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  - Total Length: 84
  - Identification: 0x4922 (18722)
  - > 0000 .... = Flags: 0x0
  - ...0 0000 0000 0000 = Fragment Offset: 0
  - Time to Live: 64
  - Protocol: ICMP (1)
  - Header Checksum: 0xae1f [validation disabled]
  - [Header checksum status: Unverified]
  - Source Address: 192.168.1.12
  - Destination Address: 192.168.1.11
  - [Stream index: 0]
- Internet Control Message Protocol
  - Type: Echo (ping) request (8)
  - Code: 0
  - Checksum: 0xfdc1 [correct]
  - [Checksum Status: Good]
  - Identifier (BE): 8777 (0x2249)
  - Identifier (LE): 18722 (0x4922)
  - Sequence Number (BE): 1 (0x0001)
  - Sequence Number (LE): 256 (0x0100)
  - [Response frame: 12]
  - > Data (56 bytes)

**Raw Data (Hex/ASCII):**

```
0000 00 50 79 66 68 00 00 50 79 66 68 01 00 00 45 00  Pyfh..P yfh..E
0010 00 54 49 22 00 00 40 01 ae 1f c0 a8 01 0c c0 a8  TI".@ .....
0020 01 0b 08 00 fd c1 22 49 00 01 08 00 0a 0b 0c 0d  .....I .....
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d  ...!"#$%&'()*+,.
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d  ./012345 6789:;<
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d  >?
0060 3e 3f
```

Рисунок 8: Анализ ARP и ICMP Echo Request/Echo Reply в Wireshark

## 2.9 Выполняю UDP-проверку

```
VPCS>  
VPCS> ping 192.168.1.11 -c 1 -2  
  
64 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=1.143 ms  
  
VPCS> |
```

Рисунок 9: Одиночный запрос к PC1 в UDP-режиме

## 2.10 Анализирую UDP Echo

| No. | Time       | Source           | Destination      | Protocol | Length | Info                                                           |
|-----|------------|------------------|------------------|----------|--------|----------------------------------------------------------------|
| 8   | 2.852283   | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                      |
| 9   | 254.399614 | Private_66:68:01 | Broadcast        | ARP      | 64     | Who has 192.168.1.11? Tell 192.168.1.12                        |
| 10  | 254.400916 | Private_66:68:00 | Private_66:68:01 | ARP      | 64     | 192.168.1.11 is at 00:50:79:66:68:00                           |
| 11  | 254.402411 | 192.168.1.12     | 192.168.1.11     | ICMP     | 98     | Echo (ping) request id=0x2249, seq=1/256, ttl=64 (reply in 12) |
| 12  | 254.403335 | 192.168.1.11     | 192.168.1.12     | ICMP     | 98     | Echo (ping) reply id=0x2249, seq=1/256, ttl=64 (request in 11) |
| 13  | 377.199915 | Private_66:68:01 | Broadcast        | ARP      | 64     | Who has 192.168.1.11? Tell 192.168.1.12                        |
| 14  | 377.200595 | Private_66:68:00 | Private_66:68:01 | ARP      | 64     | 192.168.1.11 is at 00:50:79:66:68:00                           |
| 15  | 377.202867 | 192.168.1.12     | 192.168.1.11     | ECHO     | 98     | Request                                                        |
| 16  | 377.203458 | 192.168.1.11     | 192.168.1.12     | ECHO     | 98     | Response                                                       |

[Coloring Rule String: udp]

▼ Ethernet II, Src: Private\_66:68:01 (00:50:79:66:68:01), Dst: Private\_66:68:00 (00:50:79:66:68:00)

- > Destination: Private\_66:68:00 (00:50:79:66:68:00)
- > Source: Private\_66:68:01 (00:50:79:66:68:01)
- Type: IPv4 (0x0800)
- [Stream index: 4]

▼ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11

0100 .... = Version: 4

... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x499d (18845)

> 0000 .... = Flags: 0x0

... 0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: UDP (17)

Header Checksum: 0xad94 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.12

Destination Address: 192.168.1.11

[Stream index: 0]

▼ User Datagram Protocol, Src Port: 53088, Dst Port: 7

Source Port: 53088

Destination Port: 7

Length: 64

Checksum: 0x1114 [Unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 1]

> [Timestamps]

UDP payload (56 bytes)

> Echo

```
0000 00 50 79 66 68 00 00 50 79 66 68 01 00 00 45 00  Pyfh..P yfh..E
0010 00 54 49 9d 00 00 40 11 ad 94 c0 a8 01 0c c0 a8  TI...@ .....
0020 01 0b cf 60 00 07 00 40 11 14 00 50 79 66 68 01  ...'..@ ...Pyfh
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d  .....
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d  .. 1"#$%&'()*+,.
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d  ./012345 6789:;<
0060 3e 3f                                     >?
```

Рисунок 10: Анализ UDP Echo Request и Echo Response в Wireshark

## 2.11 Выполняю TCP-проверку

```
VPCS>  
VPCS> ping 192.168.1.11 -c 1 -3  
  
Connect 7@192.168.1.11 seq=1 ttl=64 time=4.127 ms  
SendData 7@192.168.1.11 seq=1 ttl=64 time=2.397 ms  
Close 7@192.168.1.11 seq=1 ttl=64 time=2.961 ms  
  
VPCS> █
```

Рисунок 11: Проверка соединения между узлами в TCP-режиме

## 2.12 Анализирую ТСП-соединение

[illegible]

### Рисунок 12: Анализ TCP-соединения и передачи Echo-данных в Wireshark

## 2.13 Создаю сеть с маршрутизатором FRR

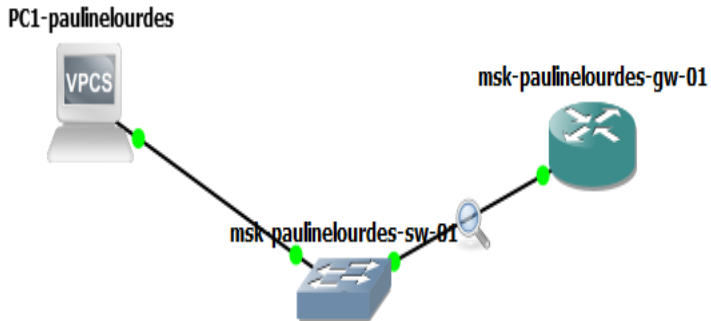
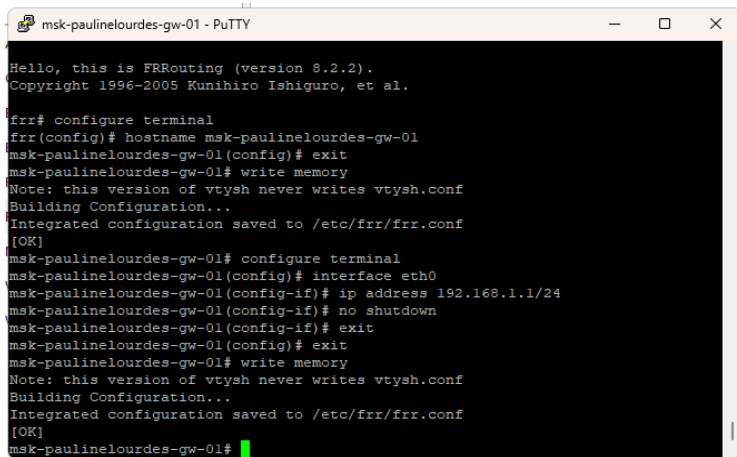


Рисунок 13: Топология сети с маршрутизатором FRR

## 2.14 Настраиваю маршрутизатор FRR

A screenshot of a PuTTY terminal window titled "msk-paulinelourdes-gw-01 - PuTTY". The terminal shows the FRRouting (version 8.2.2) prompt. The user enters "configure terminal", then "hostname msk-paulinelourdes-gw-01", then "exit", then "write memory". A note appears: "Note: this version of vtysh never writes vtysh.conf". The configuration is saved to "/etc/frr/frr.conf". The user then enters "configure terminal" again, followed by "interface eth0", "ip address 192.168.1.1/24", "no shutdown", "exit", "exit", "write memory", and another note about vtysh. The configuration is saved again to "/etc/frr/frr.conf". The terminal ends with the prompt "msk-paulinelourdes-gw-01#" and a green cursor.

```
msk-paulinelourdes-gw-01 - PuTTY

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-paulinelourdes-gw-01
msk-paulinelourdes-gw-01(config)# exit
msk-paulinelourdes-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-paulinelourdes-gw-01# configure terminal
msk-paulinelourdes-gw-01(config)# interface eth0
msk-paulinelourdes-gw-01(config-if)# ip address 192.168.1.1/24
msk-paulinelourdes-gw-01(config-if)# no shutdown
msk-paulinelourdes-gw-01(config-if)# exit
msk-paulinelourdes-gw-01(config)# exit
msk-paulinelourdes-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-paulinelourdes-gw-01#
```

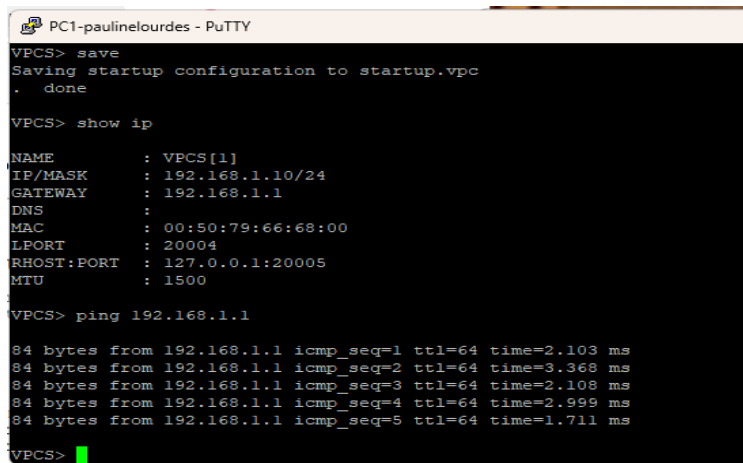
Рисунок 14: Настройка имени и IP-адреса интерфейса маршрутизатора FRR

## 2.15 Проверка конфигурации FRR

```
msk-paulinelourdes-gw-01#  
msk-paulinelourdes-gw-01# show running-config  
Building configuration...  
  
Current configuration:  
!  
frr version 8.2.2  
frr defaults traditional  
hostname frr  
hostname msk-paulinelourdes-gw-01  
service integrated-vtysh-config  
!  
interface eth0  
  ip address 192.168.1.1/24  
exit  
!  
end  
msk-paulinelourdes-gw-01# show interface brief  
Interface      Status    VRF      Addresses  
-----  
eth0            up        default  192.168.1.1/24  
eth1            down      default  
eth2            down      default  
eth3            down      default  
eth4            down      default  
eth5            down      default  
eth6            down      default  
eth7            down      default  
lo              up        default  
pimreg          up        default  
msk-paulinelourdes-gw-01#
```

Рисунок 15: Проверка текущей конфигурации и интерфейсов FRR

## 2.16 Проверяю связь PC1 с FRR



```
PC1-paulinelourdes - PuTTY
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.103 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=3.368 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.108 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.999 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.711 ms

VPCS> 
```

Рисунок 16: Проверка IP-конфигурации PC1 и связи с маршрутизатором FRR

## 2.17 Анализирую ICMP-трафик FRR

| No. | Time       | Source       | Destination  | Protocol | Length | Info                                                            |
|-----|------------|--------------|--------------|----------|--------|-----------------------------------------------------------------|
| 18  | 166.000491 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1b4b, seq=1/256, ttl=64 (request in 17)  |
| 19  | 167.003068 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1c4b, seq=2/512, ttl=64 (reply in 20)  |
| 20  | 167.004905 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1c4b, seq=2/512, ttl=64 (request in 19)  |
| 21  | 168.007355 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1d4b, seq=3/768, ttl=64 (reply in 22)  |
| 22  | 168.008865 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1d4b, seq=3/768, ttl=64 (request in 21)  |
| 23  | 169.009194 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1e4b, seq=4/1024, ttl=64 (reply in 24) |
| 24  | 169.009204 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1e4b, seq=4/1024, ttl=64 (request in 23) |
| 25  | 170.009319 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1f4b, seq=5/1280, ttl=64 (reply in 26) |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&gt; Frame 19: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0 &gt; Ethernet II, Src: Private_06:08:00 (00:50:79:06:08:00), Dst: 0c:24:9f:67:00:00 (0c:24:9f:67:00:00)   &gt; Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1     0100 .... = Version: 4     .... 0101 = Header Length: 20 bytes (5)     &gt; Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)       Total Length: 84       Identification: 0x4b1c (19228)     &gt; 0000 .... = Flags: 0x0       ...0 0000 0000 0000 = Fragment Offset: 0       Time to Live: 64       Protocol: ICMP (1)       Header Checksum: 0xac31 [validation disabled]       [Header checksum status: Unverified]       Source Address: 192.168.1.10       Destination Address: 192.168.1.1       [Stream index: 0]   &gt; Internet Control Message Protocol     Type: Echo (ping) request (8)     Code: 0     Checksum: 0x03bf [correct]     [Checksum Status: Good]     Identifier (BE): 7243 (0x1c4b)     Identifier (LE): 19228 (0x4b1c)     Sequence Number (BE): 2 (0x0002)     Sequence Number (LE): 512 (0x0200)     [Response frame: 20]   &gt; Data (56 bytes)</pre> | <pre>0000 0c 24 9f 67 00 00 00 50 79 66 68 00 00 00 45 00 \$ g...P yfh...E 0010 00 54 4b 1c 00 00 40 01 ac 31 c0 a8 01 0a c0 a8 TK...@...1.... 0020 01 01 00 00 03 bf 1c 4b 00 02 08 09 0a 0b 0c 0d -----K----- 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ..... 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ..!##\$%&amp;'()*+,. 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ./012345 6789;&lt; 0060 3e 3f &gt;?</pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Рисунок 17: Анализ ICMP-трафика между PC1 и маршрутизатором FRR

## 2.18 Создаю сеть с маршрутизатором VyOS

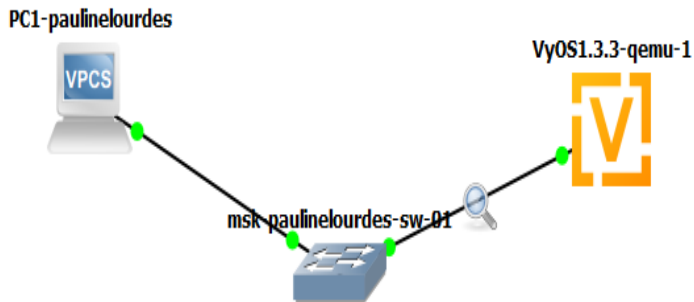
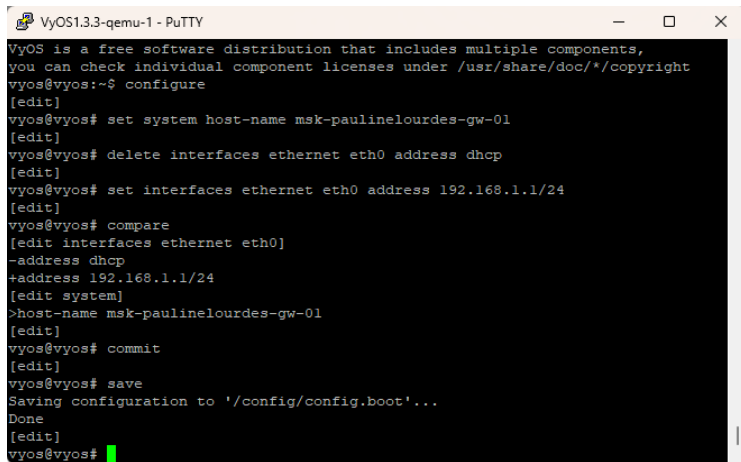


Рисунок 18: Топология сети с маршрутизатором VyOS

## 2.19 Настраиваю маршрутизатор VyOS



```
VyOS1.3.3-qemu-1 - PuTTY
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-paulinelourdes-gw-01
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
+address 192.168.1.1/24
[edit system]
>host-name msk-paulinelourdes-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

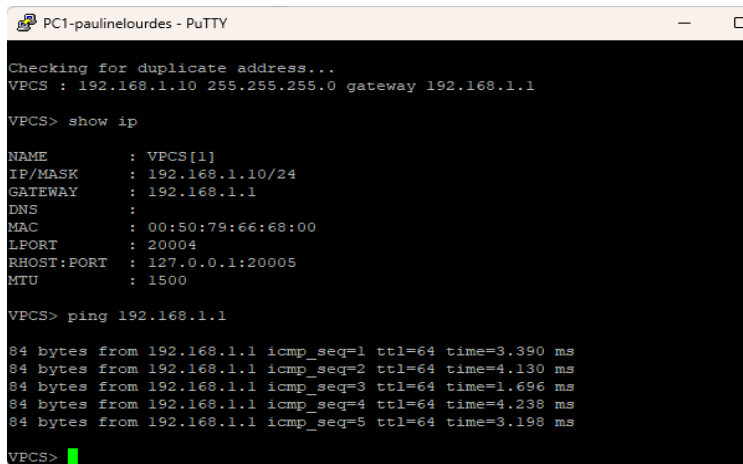
Рисунок 19: Настройка имени и адреса интерфейса eth0 маршрутизатора VyOS

## 2.20 Проверяю интерфейсы VyOS

```
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:73:04:8f:00:00
  }
  ethernet eth1 {
    hw-id 0c:73:04:8f:00:01
  }
  ethernet eth2 {
    hw-id 0c:73:04:8f:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

Рисунок 20: Проверка конфигурации интерфейсов маршрутизатора VyOS

## 2.21 Проверяю связь PC1 с VyOS



```
PC1-paulinelourdes - PuTTY

Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.390 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=4.130 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.696 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=4.238 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=3.198 ms

VPCS> 
```

Рисунок 21: Проверка IP-конфигурации PC1 и связи с маршрутизатором VyOS

## 2.22 Анализирую ICMP-трафик VyOS

| No. | Time       | Source       | Destination  | Protocol | Length | Info                                                            |
|-----|------------|--------------|--------------|----------|--------|-----------------------------------------------------------------|
| 24  | 195.232654 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0xaf4c, seq=1/256, ttl=64 (request in 23)  |
| 25  | 196.234410 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0xb04c, seq=2/512, ttl=64 (reply in 26)  |
| 26  | 196.237659 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0xb04c, seq=2/512, ttl=64 (request in 25)  |
| 27  | 197.239489 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0xb14c, seq=3/768, ttl=64 (reply in 28)  |
| 28  | 197.240660 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0xb14c, seq=3/768, ttl=64 (request in 27)  |
| 29  | 198.243572 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0xb24c, seq=4/1024, ttl=64 (reply in 30) |
| 30  | 198.246699 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0xb24c, seq=4/1024, ttl=64 (request in 29) |
| 31  | 199.248389 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0xb34c, seq=5/1280, ttl=64 (reply in 32) |

|                                                                                                      |                                                      |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| > Frame 26: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0   | 0000 00 50 79 66 68 00 0c 73 04 8f 00 00 08 00 45 00 |
| > Ethernet II, Src: 0c:73:04:8f:00:00 (0c:73:04:8f:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00) | 0010 00 54 3c 9a 00 00 40 01 ba b3 c0 a8 01 01 c0 a8 |
| > Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.10                                   | 0020 01 0a 00 00 77 bd b0 4c 00 02 06 09 0a 0b 0c 0d |
| 0100 .... = Version: 4                                                                               | 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d |
| .... 0101 = Header Length: 20 bytes (5)                                                              | 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d |
| > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                      | 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d |
| Total Length: 84                                                                                     | 0060 3e 3f                                           |
| Identification: 0x3c9a (15514)                                                                       |                                                      |
| > 000. .... = Flags: 0x0                                                                             |                                                      |
| ...0 0000 0000 0000 = Fragment Offset: 0                                                             |                                                      |
| Time to Live: 64                                                                                     |                                                      |
| Protocol: ICMP (1)                                                                                   |                                                      |
| Header checksum: 0xb0b3 [validation disabled]                                                        |                                                      |
| [Header checksum status: Unverified]                                                                 |                                                      |
| Source Address: 192.168.1.1                                                                          |                                                      |
| Destination Address: 192.168.1.10                                                                    |                                                      |
| [Stream index: 1]                                                                                    |                                                      |
| > Internet Control Message Protocol                                                                  |                                                      |
| Type: Echo (ping) reply (0)                                                                          |                                                      |
| Code: 0                                                                                              |                                                      |
| Checksum: 0x77bd [correct]                                                                           |                                                      |
| [Checksum Status: Good]                                                                              |                                                      |
| Identifier (BE): 45132 (0xb04c)                                                                      |                                                      |
| Identifier (LE): 19632 (0x4cb0)                                                                      |                                                      |
| Sequence Number (BE): 2 (0x0002)                                                                     |                                                      |
| Sequence Number (LE): 512 (0x0200)                                                                   |                                                      |
| [Request frame: 25]                                                                                  |                                                      |
| [Response time: 3,249 ms]                                                                            |                                                      |
| > Data (56 bytes)                                                                                    |                                                      |

Рисунок 22: Анализ ICMP-трафика между PC1 и маршрутизатором VyOS

### 3. 3. Выводы по проделанной работе

---

## 3.1 Вывод

В ходе лабораторной работы были построены и настроены простые сети в GNS3 на базе Ethernet-коммутатора, маршрутизатора FRR и маршрутизатора VyOS.

Для оконечных устройств VPCS была выполнена настройка IPv4-адресации в сети **192.168.1.0/24**. Связь между узлами была проверена с помощью команды **ping**.

С помощью Wireshark был выполнен захват и анализ ARP, ICMP, UDP и TCP-трафика. Были рассмотрены широковещательные ARP-запросы, ICMP Echo Request и Echo Reply, UDP Echo, а также процесс установления и завершения TCP-соединения.

Для маршрутизаторов FRR и VyOS были настроены имена устройств и IP-адреса интерфейсов **192.168.1.1/24**. Проверка показала, что оконечное устройство успешно обменивается ICMP-пакетами с каждым маршрутизатором.

В результате работы были закреплены навыки создания сетевых топологий в GNS3, настройки IP-адресации и анализа сетевого взаимодействия на разных уровнях модели